Do Genomic and Protein Language Models Share Variant Mechanisms?
Aligning Evo 2 and ESM-2 across the Central Dogma

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Abstract

A genomic language model (Evo 2) and a protein language model (ESM-2) both score human coding variants, but do they respond to the same underlying causes, or to complementary, modality-specific ones? Answering this means comparing their internal representations, which is hard: the two models differ in architecture, alphabet, length, and objective, and raw-activation similarity is dominated by gene identity and conservation. We take the change a variant induces—the paired wild-type→mutant activation delta $\Delta h_m = h_m(\text{mut}) - h_m(\text{wt})$ in each model m—and learn a shared cross-modal representation of these deltas: a contrastive linear alignment (a deep-CCA-style objective) augmented with shared and modality-private sparse dictionaries for interpretability, and a delta-crosscoder variant for model diffing. On BRCA1 saturation genome editing (2086 missense variants with deep-mutational-scanning scores) the shared representation retrieves the paired variant across modalities well above linear CCA (Recall@1 0.307 vs. 0.216), predicts function scores better than CCA and the DNA model, and—crucially for the single-gene concern—transfers to entirely unseen genes on a 15-gene ClinVar panel (gene-disjoint pathogenicity AUROC 0.886). The two models' functional directions map to nearly the same axis in the shared space (cosine 0.905 vs. 0.306 for random pairs), and the sparse codes predict loss-of-function (AUROC 0.830) and protein domain. We are equally explicit about what fails: the sparse codes do not by themselves drive the cross-modal alignment (it is deep-CCA-level), individual latents are only weakly monosemantic, and a causal probe is negative—injecting the shared functional direction does not reliably steer either model. Representational alignment across the central dogma is real and transferable, but does not (yet) imply an interchangeable causal handle. Code, activations, and harness released.

1 Introduction

Foundation models now span the central dogma: Evo 2 [Brixi et al., 2026] models DNA at single-nucleotide resolution, while ESM-2 [Lin et al., 2023] models proteins. Both assign scores to human coding variants and are used for variant effect prediction (VEP). We ask a mechanistic question about their internals: when a DNA model and a protein model both flag a missense variant, are they responding to the same cause (e.g. disruption of a conserved structural residue), or to different, modality-specific ones (splicing/codon context on the DNA side; folding/solvent exposure on the protein side)? This requires comparing representations, not output scores—hard because the models differ in architecture (StripedHyena vs. Transformer), alphabet (nucleotides vs. amino acids), length, width, and objective, and because naive activation similarity is dominated by gene identity and conservation.

Three ideas make the comparison tractable. First, the object of interest is not the absolute representation of a sequence but the change a variant induces; the paired wild-type→mutant delta in each model cancels most sequence-identity nuisance and isolates the variant's effect. Second, a variant is a natural anchor shared by the two models, so we can learn a cross-modal alignment by contrasting matched vs. mismatched deltas—without ever aligning raw sequence embeddings. Third, restricting the deep functional analysis to a single gene (BRCA1) removes the gene-identity shortcut, so any alignment must reflect within-gene variant structure; we then test transfer explicitly on many genes.

Our study is a controlled measurement, not a new architecture. The alignment component is a deep-CCA [Andrew et al., 2013]; on top of it we add shared and modality-private sparse dictionaries (in the crosscoder family [Lindsey et al., 2024]) that make the shared space interpretable and support model diffing. Contributions:

- A new question and setting: aligning the internal representations of a genomic and a protein foundation model, on paired variant-induced deltas—to our knowledge not previously studied. We name the method CentralDogma-ΔCC.
- A rigorous, multi-seed, honest evaluation on BRCA1 SGE: cross-modal retrieval and DMS prediction (vs. CCA, deep-CCA, single models, concat), residue- and domain-disjoint splits, biological probing, a shared functional axis analysis with controls, and a cross-model causal probe that returns a negative result.
- Cross-gene generalization to entirely unseen genes on a 15-gene ClinVar panel, directly addressing the single-gene concern.
- A model-diffing extension (§6): diffing base vs. LoRA-disease-fine-tuned Evo 2 with a Delta-Crosscoder [Kassem et al., 2026] shows narrow fine-tuning amplifies existing latents rather than adding new ones.

2 Related Work

Genomic/protein language models for variants. Evo 2 [Brixi et al., 2026] is a StripedHyena genomic model with strong zero-shot VEP, including on BRCA1 SGE [Findlay et al., 2018]; ESM-2 [Lin et al., 2023] is a protein masked-LM whose representations encode structure and function and top protein DMS benchmarks [Notin et al., 2023]. Prior work compares or ensembles the outputs of DNA and protein predictors [Benegas et al., 2025, Notin et al., 2023]; we instead compare their internal representations. Sparse dictionaries and interpreting biological LMs. Sparse autoencoders decompose LM activations into interpretable features [Bricken et al., 2023, Cunningham et al., 2023, Gao et al., 2024, Bussmann et al., 2024], and this has been applied separately to protein LMs (InterPLM [Simon and Zou, 2025]) and to gene/genomic LMs [Guan et al., 2025]. Interpreting a single biological LM is thus established; our contribution is to jointly decompose two models of different modalities. Crosscoders and model diffing. Crosscoders [Lindsey et al., 2024] jointly encode multiple representation spaces for cross-layer analysis and model diffing across fine-tunes [Minder et al., 2025, Kassem et al., 2026] and architectures [Jiralerspong and Bricken, 2026]—but always within one modality (text LLMs). We apply the idea across the central dogma, and reuse the Delta-Crosscoder [Kassem et al., 2026] to diff a base vs. disease-fine-tuned Evo 2 (§6). Cross-modal alignment. The component that drives our retrieval/prediction gains is a contrastive linear alignment—a deep-CCA [Andrew et al., 2013] generalisation of CCA [Hotelling, 1936]; the crosscoder adds sparse interpretable codes and the diffing extension. Aligning a genomic and a protein foundation model on paired variant deltas is, to our knowledge, new.

3 Method

Paired variant deltas. For a missense SNV we build an 8,192 bp variant-centred window and score wild-type and mutant sequences with Evo 2, taking the residual-stream activation at layer ℓ_{DNA} and the SNV position; the DNA delta is $\Delta h_{\mathrm{DNA}} = h^{\mathrm{mut}} - h^{\mathrm{wt}} \in \mathbb{R}^{4096}$. Independently we substitute the residue and take the ESM-2 hidden-state delta $\Delta h_{\mathrm{PROT}} \in \mathbb{R}^{1280}$ at the variant residue and layer ℓ_{PROT}. Deltas are standardized per dimension and projected onto the top 128 principal components (fit on the training split and frozen); variant deltas are high intrinsic dimension, so this denoising is needed for reconstruction to generalize, and freezing the basis is needed for reproducibility.

Shared alignment + sparse dictionaries. Each modality has (i) a linear alignment head $a_m = W^m_{\mathrm{align}} \Delta h_m$ and (ii) shared/private sparse encoders producing codes $z^m_s, z^m_p = \mathrm{ReLU}(\cdot)$, sparsified with BatchTopK [Bussmann et al., 2024] and decoded to reconstruct Δh_m. The objective is

$$\mathcal{L} = \underbrace{\mathrm{NMSE}_{\mathrm{DNA}} + \mathrm{NMSE}_{\mathrm{PROT}}}_{\text{reconstruction}} + \lambda_c \, \mathcal{L}_{\mathrm{InfoNCE}}(a_{\mathrm{DNA}}, a_{\mathrm{PROT}}) + \lambda_a \, \|\bar{z}^{\mathrm{DNA}}_s - \bar{z}^{\mathrm{PROT}}_s\|^2 + \lambda_o \, \mathcal{L}_{\perp}(z_s, z_p).$$

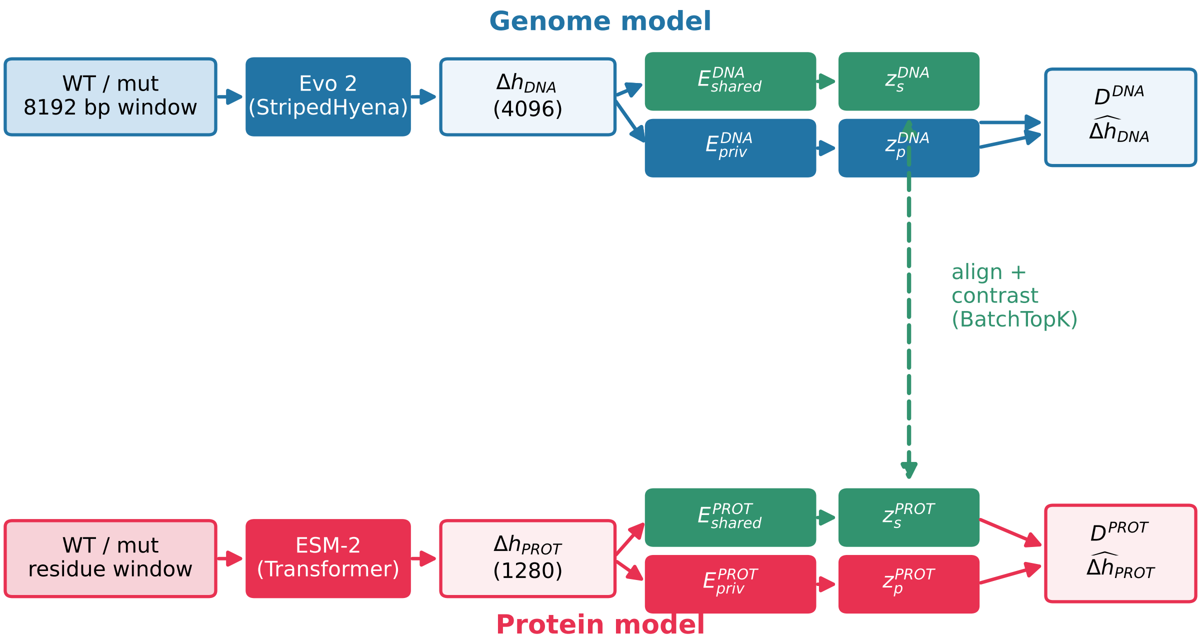

Figure 1: CentralDogma-ΔCC. For a missense variant, wild-type and mutant sequences are scored by Evo 2 (DNA) and ESM-2 (protein); the per-model activation deltas at the variant position are aligned by a contrastive linear head (used for retrieval/probing) and, in parallel, encoded into shared and modality-private sparse codes that reconstruct each modality (used for interpretability and diffing).

InfoNCE aligns the linear embeddings a_m so that row i of the DNA embedding retrieves row i of the protein embedding (all variants are in one gene, so negatives are in-gene and retrieval cannot exploit gene identity); a weak ℓ_2 term keeps the sparse shared codes consistent across modalities, and orthogonality decorrelates shared from private codes. We are explicit that the alignment a_m is a deep-CCA head; our ablation (§5) shows it, not the sparse codes, drives retrieval/prediction, while the sparse codes carry the interpretability.

Causal probe. For a direction u in the shared alignment space (e.g. the DMS-predictive direction), we map it into each model's raw activation space through that model's alignment encoder, inject $\pm\alpha u$ at the variant position during a forward pass, and re-score, comparing against a matched-norm random direction.

Model diffing (DeltaEvo). The same delta idea diffs a model against a fine-tuned copy of itself: a Delta-Crosscoder [Kassem et al., 2026] over matched $(\Delta_{\text{base}}, \Delta_{\text{ft}})$ pairs, classifying latents as shared/amplified/fine-tune-specific by their per-model decoder norms (§6).

4 Experimental setup

BRCA1 SGE [Findlay et al., 2018]: 2086 missense SNVs with continuous DMS scores, a functional class (FUNC/INT/LOF), and (partial) ClinVar labels, mapped to UniProt P38398 (all 2086 residue references validated). Variants fall in two disjoint domains (RING n=642; BRCT n=1363), giving a domain-disjoint split. For cross-gene transfer we assemble 15 genes of ClinVar missense (balanced pathogenic/benign, GRCh38 coordinates, UniProt-validated) and split by gene. Evo 2 is evo2_7b, ESM-2 is esm2_t33_650M; we select ℓ_{DNA}=blocks.24.mlp.l3, $\ell_{\text{PROT}}=ESM-2layer33$ and local-mean pooling by a ridge DMS probe on the full data. Model width: shared/private 32/96, alignment dim 64, BatchTopK k=24. Baselines: single-model Δ probe, concatenation, PLS, linear CCA, and a pure deep-CCA. All numbers are held-out, 5 seeds.

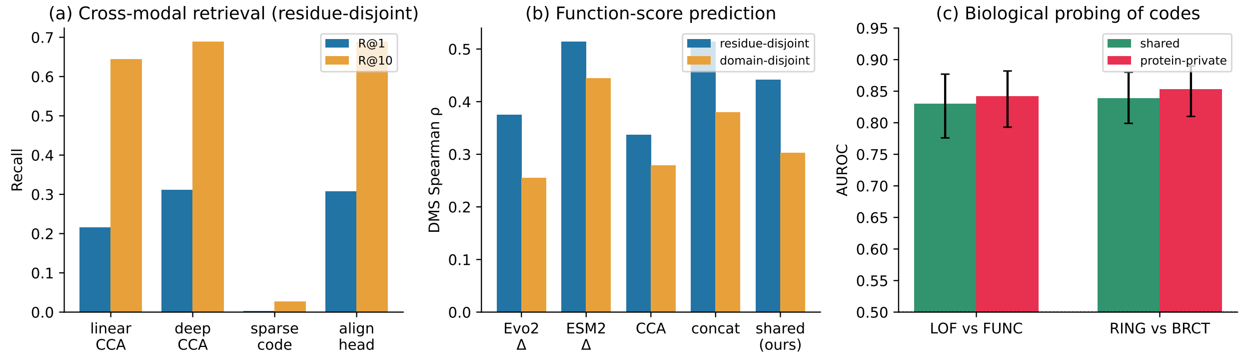

Figure 2: Results (5 seeds). (a) Cross-modal retrieval: a learned alignment (deep-CCA or our alignment head) beats linear CCA; the sparse shared code does not retrieve (it is the interpretable dictionary). (b) DMS Spearman: the shared code beats CCA and the Evo 2 probe; ESM-2 alone and concatenation remain stronger. (c) Probing the codes: the shared representation predicts loss-of-function and domain; private codes are complementary.

Table 1: Cross-modal retrieval and DMS prediction (residue-disjoint, n=366; retrieval/DMS from 5 seeds). A learned alignment (deep-CCA or our crosscoder's alignment head) beats linear CCA at both; the crosscoder's sparse shared code does not retrieve (it is the interpretable dictionary, not the alignment). ESM-2 alone / concat are the strongest DMS predictors.

Method	R@1	R@10	DMS ρ
linear CCA	0.216	0.645	0.337
deep CCA (align only)	0.311	0.689	0.464
crosscoder: sparse shared code	0.003	0.027	0.272
crosscoder: alignment head	0.307	0.689	0.442
ESM-2 Δ (probe)	–	–	0.514
concat / PLS	–	–	0.514

5 Results

The alignment beats linear CCA; the sparse code is for interpretation. Table 1. Our crosscoder's linear alignment head retrieves the paired variant at Recall@1 0.307±0.014 (5 seeds, residue-disjoint n=366), significantly above linear CCA (0.216; paired bootstrap Δ=0.104, $p < 0.001$) and matching a pure deep-CCA baseline (0.311). The sparse shared code alone does not retrieve (0.003, chance) — the cross-modal signal lives in the learned linear alignment, and the sparse codes serve interpretability (below). The same ordering holds for DMS (shared 0.442±0.003 > CCA 0.337, DNA 0.375; ESM-2 0.514 and concat 0.514 remain stronger). Reconstruction FVE is 0.72 (DNA)/0.70 (protein).

Domain-disjoint generalization. Training and evaluating only across the two disjoint domains (train BRCT, test RING; n=642), the alignment still beats CCA (Recall@1 0.106 vs. 0.081; DMS 0.303 vs. 0.279), though with substantial degradation (ESM-2 alone reaches 0.444). The method partially transfers to an unseen structural domain rather than memorizing residues.

Cross-gene generalization (multi-gene ClinVar). To test the single-gene concern directly, we assemble 15 genes of ClinVar missense variants (balanced pathogenic/benign, GRCh38 coordinates, protein references validated) and train on 10 genes, evaluating on 5 held-out genes (n=1439). Cross-modal retrieval within each unseen gene reaches Recall@1 0.066 (Recall@10 0.340), above linear CCA (0.056) and comparable to deep-CCA (0.068); the alignment thus transfers to genes never seen in training. On gene-disjoint ClinVar pathogenicity prediction, the shared code reaches AUROC 0.886, versus the Evo 2 probe 0.864, the ESM-2

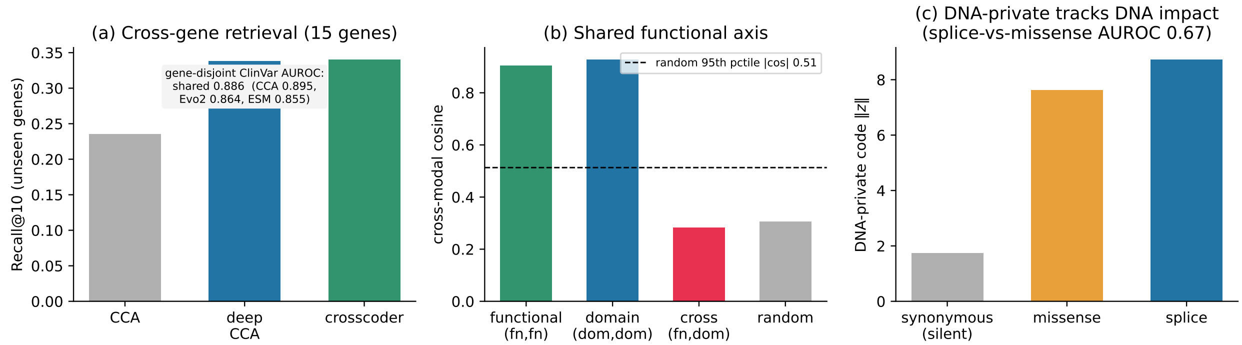

Figure 3: Generalization, shared axis, and specialization. (a) On a 15-gene ClinVar panel, the alignment transfers to unseen genes (Recall@10 within held-out genes; box: gene-disjoint pathogenicity AUROC). (b) Functional and domain directions align across modalities (cosine $\gg$ random), and are property-specific (functional vs. domain is at the random level). (c) Applied to variant classes it never trained on, the DNA-private code's activation tracks DNA-level severity: splice $>$ missense $\gg$ synonymous (silent).

probe 0.855, and CCA 0.895—the shared cross-modal representation carries transferable pathogenicity signal across genes.

Biological structure of the sparse codes. On held-out variants the shared sparse code predicts loss-of-function (AUROC 0.830, 95% CI [0.776,0.877], n=339) and domain (0.839, n=354). Modality-private codes are complementary: the protein-private code is the best domain predictor (0.853) and matches shared on LOF (0.842), consistent with ESM-2 encoding structural context. The shared code also exceeds classical single-variant VEP predictors (SIFT 0.305, CADD 0.329, phyloP 0.336; $|\rho|$ vs. DMS).

Modality-specific private codes. The private codes specialise in the expected directions (Fig. 3c). The protein-private code best predicts residue burial from the AlphaFold structure (ρ=0.41 vs. DNA-private 0.38), while the DNA-private code best predicts local sequence GC content (ρ=0.52 vs. protein-private 0.50)—a (modest) double dissociation between structure and sequence context. More strikingly, applying the frozen DNA branch to variant classes it never trained on, the DNA-private code's activation tracks DNA-level severity: $\|z\|$ is 8.7 for splice variants, 7.6 for missense, and only 1.7 for synonymous (silent) variants; it separates splice from missense at AUROC 0.67. The shared code, by contrast, fires equally on splice and missense—it encodes the cross-modal functional axis, not DNA-specific mechanism.

A shared functional axis. The direction of maximal DMS correlation, fit independently in each model's activation space, maps to nearly the same point in the alignment space (cosine 0.905), far above random direction pairs (0.306, 95th pctile $|\cos|$ = 0.513). This is property-specific: the two domain directions also align (0.927), while a functional-vs-domain pair sits at the random level (0.282). Same biological property $\Rightarrow$ same shared axis across modalities; different properties $\Rightarrow$ different axes.

Causal probing (a negative result). We test whether this alignment yields a causal handle by injecting the shared functional direction at the variant position and re-scoring (n=100 held-out variants). We tried two interventions—adding the shared functional direction, and activation-patching that removes the variant's change along that direction—and neither gives reliable control: the score shift is at best weakly correlated with the variant's true effect and indistinguishable from a matched-norm random direction (injection: ESM-2 ρ=0.131, p=0.19, Evo 2 ρ=0.009; Evo 2 patching ρ=0.24, p=0.06, specificity vs. random $\approx$ 0.003). Representational alignment does not imply an interchangeable causal handle—a caution for cross-model steering, and an open question of whether richer interventions would succeed.

6 Extension: model-diffing a disease fine-tune (DeltaEvo)

The same delta framework diffs a model against a fine-tuned copy of itself. We LoRA-fine-tune Evo 2 [Hu et al., 2022] (rank-8 adapters on the MLPs of blocks 20–26, backbone frozen) to classify BRCA1 loss-of-function from its variant-position delta activation, on the domain-disjoint split. (Evo 2's inference stack stores weights as inference tensors that cannot be differentiated; we re-materialise them to enable LoRA autograd—a small reusable recipe.) Fine-tuning improves held-out RING LOF classification from AUROC 0.562 (frozen backbone + head) to 0.665 (LoRA), and the adaptation lives in the backbone: a frozen head on the fine-tuned activations already reaches 0.614, and the mean per-variant activation change is $\|\Delta_{\text{ft}} - \Delta_{\text{base}}\| = 4.56$ (not a head-only shortcut). Training a Delta-Crosscoder on the matched pairs, it reconstructs the base→ft difference better than a standard crosscoder (ΔFVE +1.50), and its taxonomy has 246 shared and 10 amplified latents but 0 fine-tune-specific ones: narrow disease fine-tuning amplifies existing features rather than introducing new ones. The amplified latents are only weakly LOF-selective (AUROC 0.531), leaving open which reweighted features drive the gain—a preliminary but encouraging demonstration that Delta-Crosscoders can audit genomic-LM fine-tuning.

7 Discussion and limitations

We are deliberately explicit about what does and does not work. (i) Method is a setting, not a new architecture. The alignment is a deep-CCA; the sparse dictionaries add interpretability and diffing. Our ablation shows the sparse codes do not improve alignment (they retrieve at chance), so the headline retrieval/DMS gains should be read as "learned cross-modal alignment of variant deltas beats linear CCA," with the crosscoder contributing the interpretable codes and the shared-axis and diffing analyses. (ii) Assay coverage. The DMS-level analysis is on BRCA1 (to remove the gene-identity confound); we show cross-gene transfer on 15 ClinVar genes, but extending the continuous-DMS analysis to many assays (e.g. ProteinGym [Notin et al., 2023]) is the natural next step. (iii) Not a SOTA predictor. The shared code beats CCA and the DNA probe, but ESM-2 alone predicts missense DMS better; the value is interpretability, cross-modal retrieval, transfer, and the shared-axis result—not raw accuracy. (iv) Representation- vs latent-level interpretability. The shared representation predicts LOF and domain, but individual sparse latents are only weakly annotation-selective, so we report collective probes rather than claim monosemantic features. (v) Alignment $\neq$ causal exchange. Representational alignment (cosine 0.905) does not translate into a causal handle: injecting the shared functional direction is non-specific in Evo 2 and only a marginal, non-significant trend in ESM-2. This may reflect the difficulty of steering Evo 2's long-convolutional stack with a single-position edit as much as an absence of shared causal structure, and deserves dedicated study with richer interventions.

8 Conclusion

Do genomic and protein language models share variant mechanisms? At the level of representations, partly yes: a learned alignment of paired variant deltas beats linear CCA, transfers to unseen genes and domains, and places the two models' functional directions on the same axis. At the level of individual causal features, not demonstrably: sparse latents are weakly monosemantic and the shared direction is not an interchangeable causal handle. Measuring this honestly—across the central dogma, with proper controls and negative results—is, we argue, a useful step toward cross-modal mechanistic interpretability of biological foundation models.

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